Physical Chemistry CUME Nov. 8th, 2013

Entropy and the Clausius Equation

1. State the Second Law of Thermodynamics.

2. There are three variables in the Clausius Equation $dS \geq \frac{\delta Q}{T}$, please specify whether each variable is a state function or a path function, whether each variable is an intensive property or extensive property. When will the equal sign hold in the Clausius Equation?

3. Consider a system with a fixed number of particles *N* has an internal energy of *U* and a fixed volume of *V*. The temperature of the system is *T*. The entropy of the system is $S(N,V,U)$. Suppose that the energy of the system is changed quasistatically by an infinitesimal amount *dU*, keeping V and N fixed. What is the change in the entropy *dS*?

4. Suppose a hot block, at temperature T_h, is in contact with a cold block, at a temperature T_c. and $T_h > T_c$. Suppose a very small amount of heat δQ is transferred from the cold block to the hot block. Please calculate the entropy change *dS* of this process. Is *dS* positive or negative?

5. According to Boltzmann's entropy formula, $S = k_B \ln(W)$ where k_B is the Boltzmann's constant and *W* is the "number of states" available to the system. For a single gas molecule in an accessible volume *V*, the number of states is proportional to *V*. Assume a system with *N* non-interacting molecules in a piston with a volume V_L spontaneously contracts to a volume *Vs*, $V_L > V_S$. Please calculate the entropy change ΔS of the gas associated with this process. Is ΔS positive or negative?